Distance vs. displacement

02 H

DIRECTIONS · Fill all four boxes for each term. Use a number-line sketch in at least one box.

Distance

DEFINITION · MY WORDS

HOW TO FIND IT

Σ |each leg| = ____

EXAMPLES

NON-EXAMPLES

Displacement

DEFINITION · MY WORDS

FORMULA + SIGN

Δx = ____ − ____

SKETCH AN EXAMPLE

HOW IT DIFFERS FROM DISTANCE

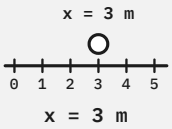
One sentence using both terms: _____

K1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: where and how far

Language goal: I can **compare** distance and displacement.



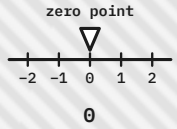
position x

posición · posição

where it is, from zero

my language: _____

use it: _____



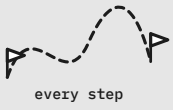
origin

origen · origem

the zero point

my language: _____

use it: _____



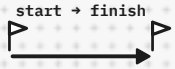
distance

distancia · distância

the whole path length

my language: _____

use it: _____



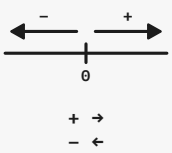
displacement

desplazamiento · deslocamento

change in position, with direction

my language: _____

use it: _____



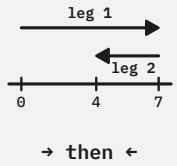
direction

dirección · direção

which way: + or −

my language: _____

use it: _____



leg

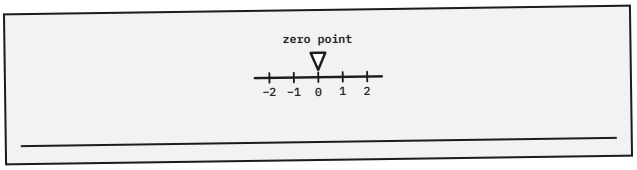
tramo · trecho

one part of a trip

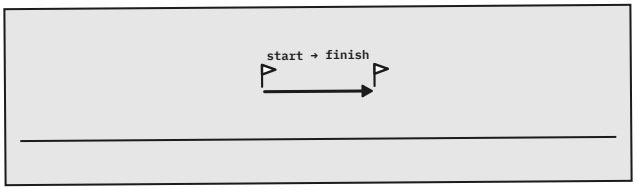
my language: _____

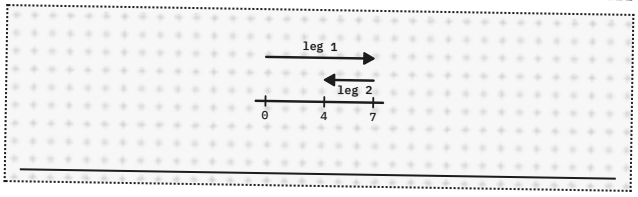
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

K1 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Fix the group chat

02 R

DIRECTIONS · Reply in the empty bubbles. Show the math on the right.

Hallway walk crew

ELI

walked 4 m to the door and back. my displacement = 8 m

YOU ①

MAZ

my locker is at x = −3 m. negative means I moved left, right?

YOU ②

NADIA

I went −3 m → +2 m → −1 m. distance?

YOU ③

ELI

and her displacement??

YOU ④

SHOW YOUR RECEIPTS

① distance vs. displacement

② position vs. motion

③ Σ |each leg|

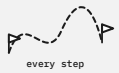
④ Δx = x_f − x_i

K1 1 2 3

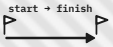
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it

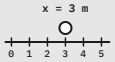
Language goal: I can **compare** distance and displacement.



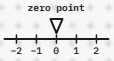
distance
distancia · distância
all of the path



displacement
desplazamiento · deslocamento
start → finish



position
posición · posição
where it is



origin
origen · origem
the zero point

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: distance · displacement · position · zero · direction

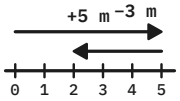
My ____ tells where I am from a ____ point. ____ counts every step I walk. ____ compares only start and finish, and it has a ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



I walked ____ m in total, so my distance is ____ m.
I ended ____ m from my start, so my displacement is ____ m.

3 · SOLVE WITH THE STEPS



Walk from 0 m to 5 m, then back to 2 m.

STEP 1 · ADD THE LEGS
____ + ____ = ____ m

STEP 2 · END − START
____ − ____ = ____ m

STEP 3 · SAY IT
"Distance ____ m, displacement ____ m."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ Distance and displacement are different because ____ .
☐ My displacement is zero because ____ .
☐ First I was at ____, then ____, so ____ .

I said it ☐ · I wrote it ☐

K1 1 2 3

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UNIT 1 · DAY 2 · K1 · CASE FILE

02 T

The missing hall pass

Hallway cameras logged a student's position (door = 0 m, east = +). Reconstruct the route.

EXHIBIT A · CAMERA LOG

TIME	POSITION	LOCATION
8:01	0 m	door
8:02	+12 m	water fountain
8:04	−5 m	locker
8:05	+3 m	classroom

EXHIBIT B · DRAW THE ROUTE ON A NUMBER LINE (−6 to +14 m)

ANALYSIS 1 · DISTANCE

Add every leg: |12 − 0| + |−5 − 12| + |3 − (−5)|

ANALYSIS 2 · DISPLACEMENT

Δx = final − start

WITNESS STATEMENT · TRUE OR FALSE?

"She only went 3 m — that's her displacement, so that's how far she walked."

THE VERDICT

Distance ____ m · displacement ____ m. They differ because ____

K1 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Position, distance, or displacement?

02 N

DIRECTIONS · Write each card's letter in its column.

A "3 m east of the door"	**B** 400 m around a track	**C** "ended 2 m left of start"	**D** never negative
E $\Delta x = x_f - x_i$	**F** $x = -2$ m	**G** counts every leg	**H** can be 0 after a long trip
I needs an origin	**J** a scalar	**K** a vector	**L** a signed coordinate

POSITION	DISTANCE	DISPLACEMENT
where it is	total path	start → finish

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

K1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 2 · K1 · COMIC STRIP

Comic: the long way vs. the shortcut

02 CS

YOUR READER

A younger cousin who thinks distance and displacement mean the same thing.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◯ PENCIL — The wandering path, dashed and labeled **distance** = ____ **m**
- ☐ ◌ BLUE — One straight arrow from start to end, labeled **displacement** = ____ **m**
- ☐ ◯ PENCIL — Units (m) on every number

1 · START *A pet (or person) at its starting spot. Mark the start.*

CAPTION _____

2 · THE WANDER *It wanders all over. Draw its whole path.*

CAPTION _____

3 · THE END *Where it ends up. Compare the path to a straight line.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **distance · displacement · start · end**

K1 1 2 3

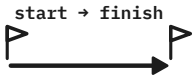
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · displacement

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Displacement · desplazamiento (ES) · deslocamento (PT)



TODAY'S CASE

A cart starts at 2 m, passes 6 m, and finishes at 4 m. Right is positive.

01 · SKETCH AND LABEL

Trace the two legs in pencil. Label the three positions in blue.

02 · CHANGE THE PICTURE

On a second line draw one blue arrow from start to finish. Write the distance beside the path and the displacement beside the arrow.

Annotate your sketches. Circle the mark that best shows displacement. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The path is longer than the displacement because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else